

One-Third Country Narrative: Structured Pedagogy (Schooling)

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Introduction

Structured pedagogy is a comprehensive educational framework/teaching and learning approach that combines an intentionally designed system of educational resources, instructional expertise, and academic learning time with a detailed, evidence-based curriculum. Its primary goal is to systematically foster the cognitive development of every student, building the mental architecture required for lifelong learning. This narrative explores structured pedagogy across three different countries that represent population income groups: the Bottom-Third Country (those with household incomes less than \$64,400), the Middle-Third Country (those with incomes less than \$130,000), and the Top-Third Country.

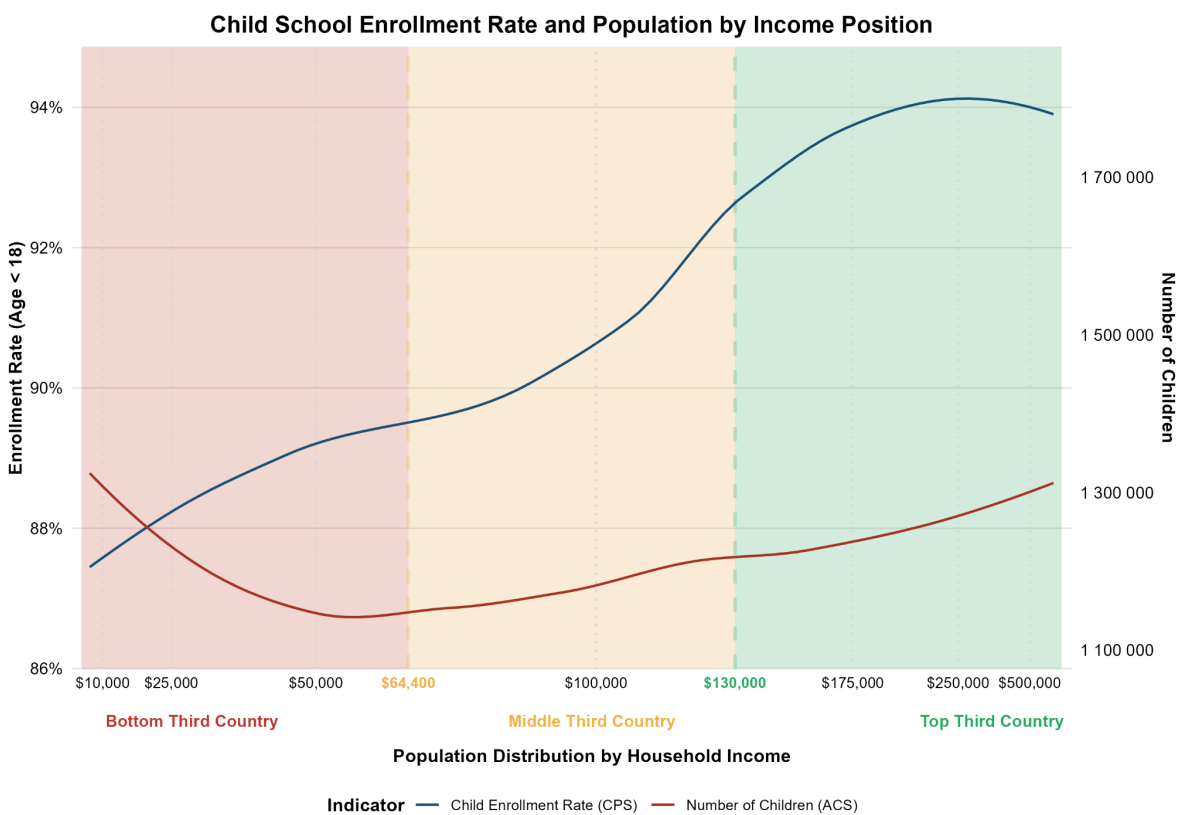
Structured pedagogy rests on three key components:

- **Academic learning time:** The time students spend actively and successfully engaged in developmentally appropriate academic content; measured through attendance, engagement, and classroom stability, which are essential for knowledge consolidation.
- **Instruction quality:** The effectiveness of pedagogical practices in fostering cognitive growth and affective outcomes such as motivation and belonging, enabling the development of higher-order thinking.

- **Educational resources:** The supports and services beyond direct instruction that make learning possible, including materials, technology, tutoring, counseling, and nutrition, which reduce cognitive load and provide the tools for learning.

Structured pedagogy directly shapes cognitive development. In order to receive the benefits in structured pedagogy, they must be enrolled in school. Yet 1-in-10 children in the Bottom-Third and most of the Middle-Third Countries are not enrolled in any formal schooling, compared to just under 1-in-20 students for the Top-Third. A slim majority of children live in the Top-Third, 24.9 million, compared to 23.8 million for the Bottom-Third and 23.7 million for the Middle-Third. Yet by far the largest concentration exist at the poorest section of the Bottom-Third. Two million children live in households with annual incomes less than \$11,000. These children are also the least likely to enroll in formal schooling, resulting in the absence of a key driver of cognitive development. Figure 1 compares enrollment rates with children populations for the different countries.

Figure 1: School Enrollment and Population Distribution (Under 18; 2023)^{1,2,3,4}



¹ Ruggles et al., 2022

² Ruggles et al., 2023a

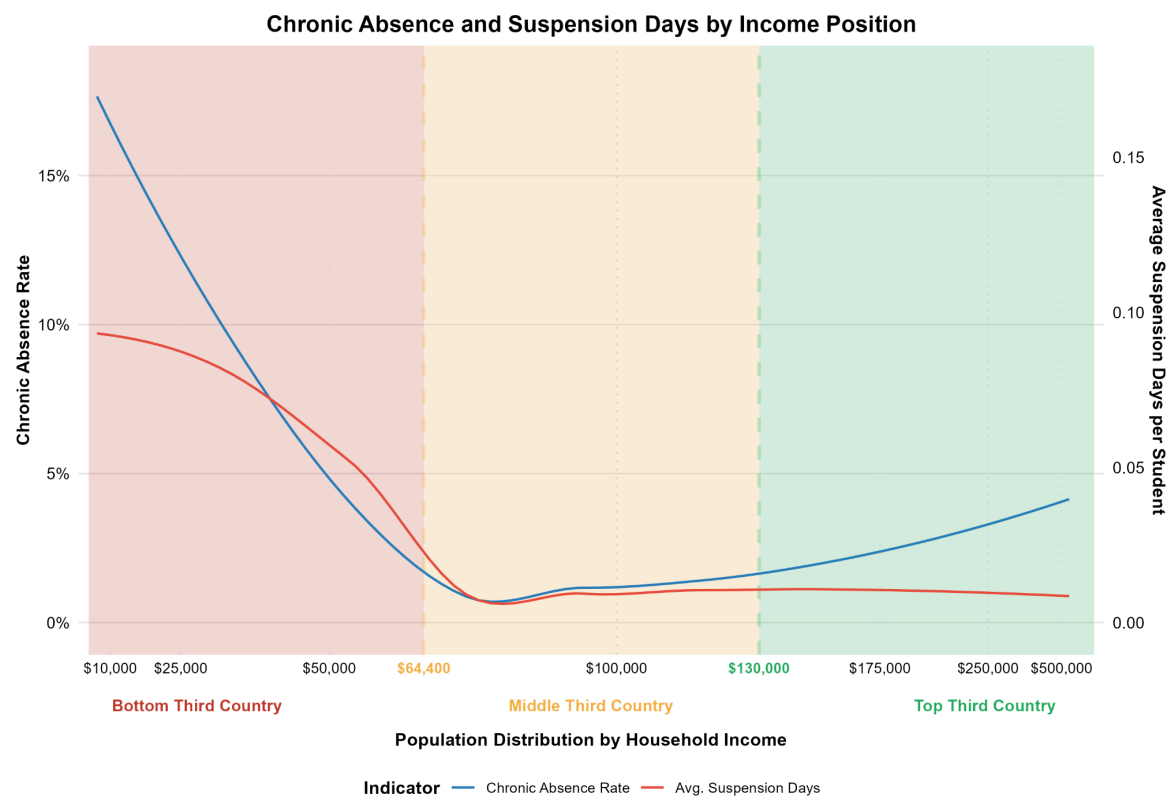
³ Ruggles et al., 2023b

⁴ Flood et al., 2023

Academic Learning Time

For students in the Bottom-Third Country, educational deficits begin with poor attendance. As shown in Figure 1, rates of chronic absenteeism and the average suspension days per students are substantially higher than the other countries, creating significant deficits in the engagement of sequential learning. This loss is compounded within the school day; high-poverty schools lose significant instructional time to classroom disruptions and external stressors, estimated at 18 minutes daily and over 12% of the school year.⁵ The classroom environment is often chaotic, forcing teachers to prioritize management over instruction. Consequently, student engagement is low, with inconsistent participation and attention. This environment fuels poor achievement rates, as students enter high school years behind their peers and face an 11.6% dropout rate.⁶ The long-term implications are severe: inconsistent engagement slows cognitive development, particularly in critical thinking and problem-solving. This system perpetuates a cycle of limited access to higher education and well-paying jobs, reinforcing generational inequality.

Figure 2: Chronic Absenteeism and Suspensions (2017)^{7,8,9,10,11}



⁵ Rodgers and Mirra, 2014

⁶ American Psychological Association, n.d.

⁷ Common Core of Data, 2017

⁸ Civil Rights Data Collection, 2017

⁹ National Historical Geographic Information System, 2017

¹⁰ Ruggles et al., 2023

¹¹ Mason et al., 2023

In the Middle-Third Country, academic learning time is generally consistent. Attendance rates are steady with low truancy, allowing for uninterrupted scheduled instructional time. The classroom environment is structured and responsive, with teachers providing support when students seek help. This fosters a climate where students show moderate engagement, regularly participating and staying on task. However, this model is often reactive, addressing weaknesses as they appear rather than proactively pushing students beyond baseline competency. Achievement rates reflect this stability, aligning closely with national averages, with a dropout rate of 3.6%.¹² While these students avoid major skill gaps, this focus on adequacy can limit aspirations. The long-term implication is a learning structure that ensures foundational competence but may not identify hidden potential or foster the skills needed for exceptional achievement, potentially capping future academic and career trajectories.

The educational system in the Top-Third Country is defined by optimized academic learning time. Attendance is consistently high, reflecting a culture that views education as essential for success. Scheduled instructional time is expanded beyond the standard day through enrichment programs, creating a 200-hour annual learning gap over the low-income households.¹³ The classroom environment is highly effective, characterized by low student-to-teacher ratios, proactive management, and individualized attention. This structure cultivates high student engagement, with sustained focus and consistent participation. As a result, achievement rates are uniformly high; students are eight times more likely to earn a bachelor's degree, and the dropout rate is only 2.8%.¹⁴ This system creates a positive feedback loop where success fuels motivation. Cognitively, the uninterrupted, high-quality instruction enhances memory, critical thinking, and higher-order reasoning, preparing students for future academic and professional success.

Instruction Quality

Instructional quality in the Bottom-Third Country is critically compromised by reliance on emergency-accredited teachers who often lack robust terminal pedagogical training and deep content expertise,¹⁵ immediately widening the knowledge gap,¹⁶ especially in STEM. There is a systemic absence of institutionalized professional development opportunities, forcing teachers to rely on external funding. This perpetuates outdated pedagogical practices and low teacher experience, significantly compromising the skill-acquisition of pupils.¹⁷ Class size is severely overcrowded, forcing instructors to manage logistics rather than teach higher-order skills.¹⁸ This limits the personalized attention required for effective learning and leads to slowed cognitive development and chronic absenteeism. High administrator turnover creates constant instability, as teachers are perpetually readjusting to new management.¹⁹ This instability results in

¹² Bradley, 2022

¹³ Kraft and Nocivoff, 2022

¹⁴ American Psychological Association, n.d.

¹⁵ Chambers, 2025

¹⁶ Taie, 2019

¹⁷ Goe, 2007

¹⁸ United Federation of Teachers, 2023

¹⁹ Superville, 2019

fragmented instruction that critically harms teacher retention and substantially depresses a student's future academic performance in school and life.²⁰

In the Middle-Third Country, instructional quality is inconsistent due to regional disparities and low baseline requirements for teacher training. Professional development opportunities are geographically concentrated, then universally systematic, leading to uneven teacher growth and inconsistent adoption of structured pedagogy.²¹ While moderate teacher experience is common, geographic variability in content expertise necessitates that teachers stretch limited resources and curriculum.²² Class size is a significant challenge, particularly in rural areas, where overcrowding is common.²³ Larger classes shift instructional focus from higher-order skills and critical thinking to basic comprehension.²⁴ This creates a meager scaffolding of subject matter and diminishes the crucial student-teacher relationship. These inequities are compounded by varying administrator quality and high turnover rates, which lead to inconsistent student outcomes and impede the sustained cognitive growth necessary for future success and workforce competitiveness.

Students in the Top-Third Country benefit from highly qualified, stable faculty. Instructors are far more likely to have terminal degrees, higher teaching certifications and, deep content expertise, enabling teachers to deliver rigorous and coherent instruction.²⁵ Professional development opportunities are institutionalized and sustained within these district systems, which allows for continuous refinement of communication and practice. This produces a stable workforce with high teacher experience, creating a self-reinforcing loop of instructional expertise and long-term, cumulative cognitive gains across all grade levels. Lower class sizes allow instructors to further logical tasks and focus entirely on higher-order skills, providing the frequent, personalized attention and rapid feedback essential for success.²⁶ Professional, stable administrator quality reinforces instructional coherence and prevents high turnover, optimizing the development of complex cognitive skills for successful academic and life trajectories.

Educational Resources

For students in the Bottom-Third Country, the school day is defined by a cycle of disadvantage. Facilities are often aging and poorly maintained due to a massive national funding gap,²⁷ creating a stressful learning environment. This is compounded by poor nutrition, as students have limited access to fresh, healthy meals, which hinders their ability to focus.²⁸ A critical shortage of resources defines their education. Instruction is often delivered by inexperienced

²⁰ Learning Policy Institute, 2017

²¹ The Commonwealth Institute, 2017

²² Baker et al., 2016

²³ Stovall and Dolan, 2011

²⁴ Baker et al., 2016

²⁵ Germuth, 2016

²⁶ Hutson, 2020

²⁷ International WELL Building Institute, 2021

²⁸ The Lancet Regional Health–Americas, 2022

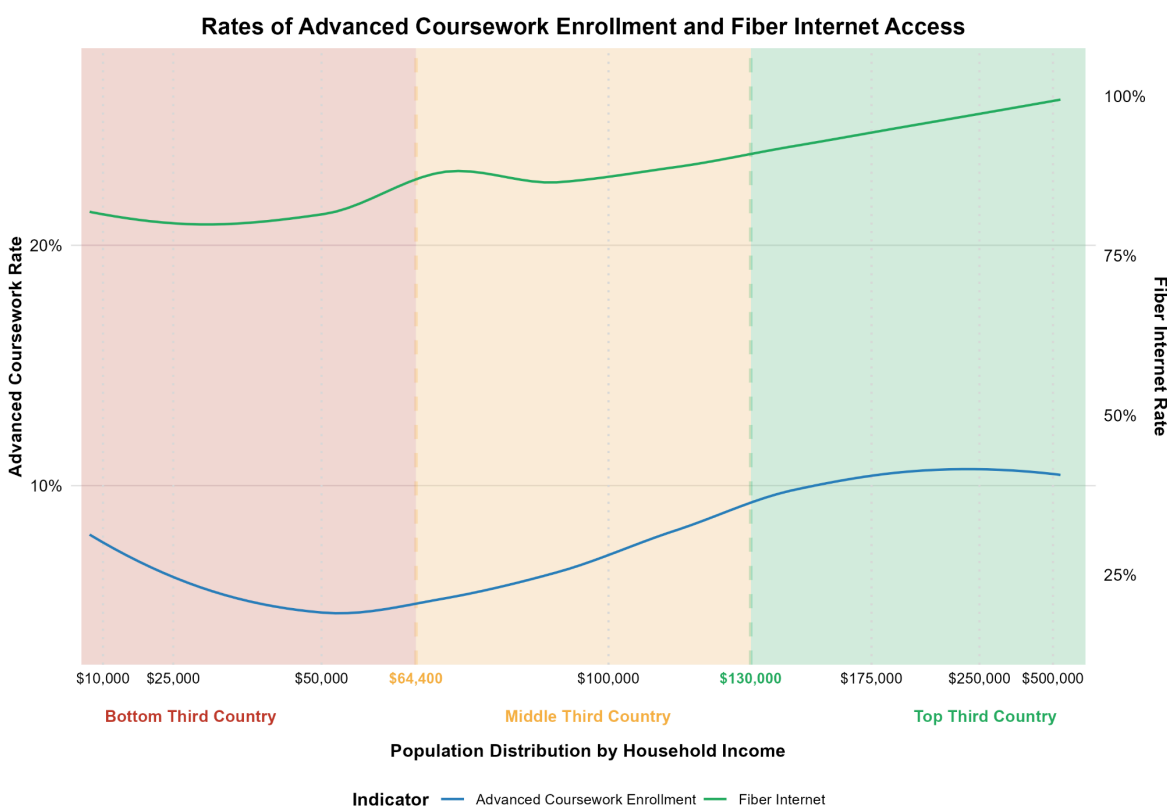
teachers,²⁹ and a severe lack of support staff means students have little access to guidance. Instructional resources are scarce, with students using outdated textbooks and classrooms lacking the digital tools and wi-fi access necessary to complete assignments. Students are also less likely to be enrolled in advanced coursework, including International Baccalaureate (IB) programs, Advanced Placement (AP) programs, and gifted and talented programs. Figure 3 showcases these inequities. Advanced coursework enrollment rates remain below 10% for the Bottom-Third and Middle-Third Countries, while only about 17-in-20 schools in the Bottom-Third are connected to fiber internet compared to 19-in-20 for the Top-Third.

Additionally, student-to-counselor ratios far exceed recommended levels,³⁰ resulting in academic and emotional needs go unmet. Beyond the classroom, few extracurricular activities are offered, denying students crucial opportunities for social and personal development. This environment of scarcity severely hinders students' cognitive development by limiting their access to critical thinking and problem-solving. Ultimately, this system perpetuates a cycle of disadvantage that restricts future access to higher education, well-paying jobs, and social mobility.

²⁹ García & Weiss, 2019

³⁰ American School Counselor Association, n.d.

Figure 3: Rates of Advanced Coursework Enrollment (2017) and Fiber Internet Access (2020)^{31,32,33,34,35,36}



For students in the Middle-Third Country, the environment is functional but lacks the resources for excellence.³⁷ Facilities are safe but aging, leading to minor but persistent disruptions that can affect student morale. In the cafeteria, the nutrition program meets federal guidelines but its reliance on processed foods often causes a post-lunch slump that hampers student focus.³⁸ This theme of adequacy extends to academics. While basic instructional resources are available,³⁹ the school's dated digital tools and library collections limit opportunities for advanced inquiry. Similarly, extracurricular activities are available but are often limited to traditional sports, creating an "opportunity gap" that narrows future pathways for students with other interests.⁴⁰ The most critical restriction is a lack of sufficient support staff. With high student-to-counselor ratios, providing proactive mental health care is nearly impossible,⁴¹ forcing a crisis-response model. The overarching implication is a "potential gap." Access to core resources prevents

³¹ Common Core of Data, 2017

³² Civil Rights Data Collection, 2017

³³ Civil Rights Data Collection, 2020

³⁴ National Historical Geographic Information System, 2017

³⁵ Ruggles et al., 2023

³⁶ Mason et al., 2023

³⁷ National Center for Education Statistics, 2014

³⁸ Food Research & Action Center, n.d.

³⁹ Scholastic, n.d.

⁴⁰ Hussar & Bailey, 2019

⁴¹ García & Han, 2019

outright failure, but the inconsistencies across the board mean the path to competence is clear, while the tools needed to achieve excellence remain just out of reach.

Schools in the Top-Third Country provide an environment where resources converge to maximize learning. Students have consistent access to healthy nutrition, ensuring they arrive at school ready to concentrate, with improved focus and memory.⁴² The facilities are modern, safe, and technologically advanced, supported by reliable funding that also helps attract and retain experienced teachers.⁴³ Abundant and up-to-date instructional resources,⁴⁴ from digital platforms to advanced textbooks, allow teachers to deliver rigorous, high-quality instruction;⁴⁵ twice as much time on grade appropriate assignment and five times as much time with strong instruction compared to low-income.⁴⁶ Students are supported by a robust network of support staff including counselors,⁴⁷ and teacher assistants who provide individualized academic and emotional guidance.⁴⁸ A wide array of extracurricular activities extends learning beyond the classroom,⁴⁹ fostering teamwork, discipline, and creativity. These compounded advantages create an uninterrupted and enriched learning experience. This system not only accelerates mastery of core skills but also develops resilience and higher-order reasoning, systematically widening the gap between these students and their peers in the lower tiers.

Conclusion

Structured pedagogy in the nation's schools is ranked by socioeconomic status, creating an abyss of educational opportunity. The framework's three core components—academic learning time, instruction quality, and educational resources—do not function in isolation. Rather, they operate as an interconnected system that either propels students toward advanced cognitive development or perpetuates a cycle of disadvantage.

For students in the Bottom-Third Country, the educational experience is defined by compounding deficits. Chronic absenteeism, compromised instruction from unstable and underqualified faculty, and a scarcity of essential resources create an environment where cognitive development is consistently interrupted. This systemic failure perpetuates a cycle of limited access to higher education and upward mobility, reinforcing generational inequality.

Those in the Middle-Third Country experience a functional but limiting educational model. While these students benefit from consistent academic learning time and adequate resources, the pedagogical approach is often reactive, geared toward ensuring baseline competency rather than fostering exceptional achievement. This creates a potential gap, where the tools for

⁴² Springer et al., 2015

⁴³ Mathewson, 2020

⁴⁴ Chingos & Whitehurst, 2010

⁴⁵ Kane, 2016

⁴⁶ Steiner, 2024

⁴⁷ Lapan et al., 2018

⁴⁸ Hemelt, 2021

⁴⁹ Snellman, 2015

foundational competence are present, but the supports required to achieve excellence remain out of reach, potentially capping future academic and career trajectories.

Students in the Top-Third Country benefit from a system where all components of structured pedagogy are maximized and mutually reinforcing. Uninterrupted learning time, highly qualified and stable faculty, and an abundance of modern resources create a positive feedback loop that accelerates mastery and cultivates critical thinking. This system not only ensures academic success but systematically widens the achievement gap.

Ultimately, the disparities in structured pedagogy reveal more than just differences in degree; they represent fundamentally separate and unequal educational systems operating in parallel. This reality systematically widens the achievement gap, reinforces generational inequality, and ensures that a child's socioeconomic status remains a powerful predictor of their life's opportunities.

Appendix A: Component Tables

Tables A1–A3 served as the framework for this narrative. They were created using the following process:

1. **Define:** Contextual definitions were created for the three components of structured pedagogy: *academic learning time*, *instructional quality*, and *educational resources*.
2. **Identify measures:** An initial literature review was done to identify 4–6 measures that were the most significant in evaluating each component.
3. **Analyze measures accross countries:** A secondary literature review explored how each measure different accross the Bottom-, Middle-, and Top-Third Countries, culminating in the following tables.

Table A1: Academic Learning Time Research Findings

Measure	Botton-Third Country	Middle-Third Country	Top-Third Country
Attendance rate: The extent to which students consistently attend school, including daily attendance, dropout, and graduation rates.	Students are more likely to experience chronic absenteeism. ⁵⁰	Students tend to have steady attendance, with consistent routines throughout the school year. ⁵¹	Students experience steady attendance with fewer disruptions and strong stability across grade levels. ⁵²
Allocated/scheduled instructional time: Total number of instructional hours students are scheduled to receive during the school year.	Instructional time is reduced by about 30 minutes daily due to frequent disruptions. ⁵³	Instructional time is generally consistent, with minor variations across schools. ⁵⁴	Instructional time is slightly shorter but better preserved due to fewer absences and disruptions. ⁵⁵
Classroom environment/management: Quality of classroom organization, teacher practices, and student-teacher	Students are less likely to seek help or participate actively, limiting opportunities for interaction and reducing classroom responsiveness. ⁵⁶	Students tend to seek more help in class, which leads to increased support and teacher interaction, shaping a more responsive climate. ⁵⁷	Students receive frequent individualized attention and benefit from proactive teacher interaction,

⁵⁰ Oregon Department of Education, 2015

⁵¹ Gottfried, 2014

⁵² Gennetian et al., 2018

⁵³ Kurtz, 2014

⁵⁴ Kraft and Novicoff, 2025

⁵⁵ Kraft and Novicoff, 2010

⁵⁶ Boyce et al., 2012

⁵⁷ Thapa, 2013

Measure	Bottom-Third Country	Middle-Third Country	Top-Third Country
interactions that support a positive, focused learning climate.			fostering a highly responsive and structured classroom climate. ⁵⁸
Achievement rate: Measures how often students are correct or achieve mastery during the time they are actively engaged in school.	Students show lower achievement, with fewer academic supports and opportunities. ⁵⁹ (Wyner et al., 2007).	Students show moderately higher achievement with better resources and support. ⁶⁰	Students tend to achieve more, supported by early advantages and well-resourced instruction. ⁶¹
Engagement rate: The proportion of instructional time during which students are actively on task, through visible participation or focused attention.	Students show low to moderate engagement, with less consistent participation and reduced attentiveness during classroom activities. ⁶²	Students show moderate engagement, regularly participating and staying on task for most of the lesson. ⁶³	Most students remain actively on task, with sustained attention and consistent participation. ⁶⁴

⁵⁸ Boyce et al, 2012

⁵⁹ Wyner, Bridgeland, & Dilulio, 2007

⁶⁰ Fahle, 2021

⁶¹ American Psychological Association, 2017

⁶² Kelly, 2008

⁶³ Jensen, 2013

⁶⁴ Bae & Lai, 2012

Table A2: Instruction Quality Research Findings

Measure	Bottom-Third Country	Middle-Third Country	High-Income Country
Terminal pedagogical training: On average, board-certified, licensed, and more educated teachers produce more positive student outcomes.	High reliance on teachers with temporary or emergency credentials and less specialized training. ⁶⁵	A positive relationship was found between a 10% salary increase and math and reading scores for the middle 50% SES districts. Increased wages correlate with a higher proportion of teachers graduating from selective colleges and possessing subject-matter qualifications. ⁶⁶	Highly-credentialed teachers are more likely to be recruited by higher-income school districts and to join the staffs of schools that provide them with better support and working conditions and more choices of grades and subjects to teach. ⁶⁷
Professional development opportunities: Inadequate pay in low-income districts limits access to professional development and instructional quality.	Access is limited and fragmented, often dependent on external grants or donor programs. This restricts teachers' ability to update skills and adopt best practices. Teachers in less affluent districts often have reduced access to high-quality PD and are less likely to benefit from learning communities and coaching efforts. ⁶⁸	Professional development is more available but varies in quality and consistency. State and district initiatives seek to address these gaps. ⁶⁹	Regular, well-funded, and institutionalized professional development supports continuous instructional improvement and teacher retention. ⁷⁰
Teacher experience: More experienced teachers tend to be more effective	Schools are more likely to have novice and out-of-field teachers, resulting in less instructional stability and expertise. ⁷¹	Teacher experience is moderate; some schools struggle with retention, affecting instructional consistency. ⁷²	Teachers are generally experienced, contributing to higher instructional quality and better student outcomes. ⁷³

⁶⁵ Alix-Garcia et al., 2013

⁶⁶ Garcia & Han, 2019

⁶⁷ Alix-Garcia et al., 2013

⁶⁸ Carver-Thomas & Darling-Hammond, 2017

⁶⁹ Darling-Hammond, 2017

⁷⁰ Darling-Hammond, 2017

⁷¹ Alix-Garcia et al., 2013

⁷² Caldass & Walberg, 2011

⁷³ McEwin & Dickinson, 1995

Measure	Bottom-Third Country	Middle-Third Country	High-Income Country
Content expertise: Teachers with strong subject-matter knowledge are better equipped to deliver rigorous, coherent instruction.	Many teachers teach outside their subject area, especially STEM; linked to lower achievement. ⁷⁴	Content alignment is improving but remains inconsistent, leading to varied student achievement. ⁷⁵	Strong alignment between teacher qualifications and subjects taught ensures high-quality instruction. ⁷⁶
Class size: Smaller class sizes enable more individualized instruction, faster feedback, and stronger student–teacher relationships. Teachers in smaller classes can spend more time on higher-order skills rather than classroom management or logistical tasks.	Larger, overcrowded classes common; impedes individualized instruction. ⁷⁷	Moderate class sizes but often above optimal. ⁷⁸	Smaller classes allow individualized attention and differentiated instruction. ⁷⁹
Administrator quality: Effective principals shape instructional quality by: hiring and retaining strong teachers, providing feedback and supporting professional growth, and creating a collaborative school culture. Weak leadership often leads to high turnover and fragmented instruction.	Less experienced, lower-rated principals; high turnover undermines school culture ⁸⁰	Quality varies; some schools benefit from strong, reform-oriented leaders, while others do not. ⁸¹	Administrators are typically professionalized, stable, and effective, fostering positive school culture and instructional improvement. ⁸²

⁷⁴ Alix-Garcia et al., 2013

⁷⁵ Goldhaber et al., 2007

⁷⁶ McEwin & Dickinson, 1995

⁷⁷ Jorgensen, 2023

⁷⁸ Massachusetts Teachers Association, n.d.

⁷⁹ Massachusetts Teachers Association, n.d.

⁸⁰ Superville, 2019

⁸¹ Jacobson et al., 2005

⁸² Jacobson et al., 2005

Table A3: Educational Resources

Measure	Bottom-Third Country	Middle-Third Country	High-Income Country
Nutrition: The availability and quality of diverse, nutritious food options that support student growth, health, and development.	Students have limited access to fresh, healthy meals, with few nutritious options available in schools or nearby communities. ⁸³	Students have moderate access to healthy food options, with some variety available in schools and neighborhoods. ⁸⁴	Students have consistent access to a wide range of nutritious meals and high-quality ingredients from both schools and community sources. ^{85 86}
Facilities: The physical condition of the school's infrastructure and learning environment.	Schools operate in aging buildings with frequent maintenance issues, such as poor ventilation, outdated systems, or unsafe conditions. ⁸⁷	Schools have reasonably maintained facilities, with ongoing improvements and occasional infrastructure upgrades.	Schools maintain high-quality facilities, with modern buildings, regular upgrades, and well-resourced environments. ⁸⁸
Extracurricular activities: School-sponsored programs outside of the standard academic curriculum, such as sports, service clubs, and academic clubs.	Schools offer few extracurricular options, and student participation is limited. ⁸⁹	Schools provide a mix of academic and non-academic extracurriculars, with moderate student involvement. ⁹⁰	Schools offer a wide range of extracurricular programs, with high levels of participation and additional enrichment opportunities. ⁹¹
Support staff: School personnel, including librarians, paraeducators, and teaching assistants, who provide essential services that support student achievement, instruction, and mental health.	Schools have limited support staff, and essential roles like counselors and librarians are often unfilled. ^{92 93}	Schools have moderate staffing, with access to key support roles such as aides and mental health professionals	Schools are fully staffed with a wide range of support personnel, including librarians, paraeducators, and counselors. ⁹⁴

⁸³ Springer et al., 2015

⁸⁴ Sanchez-Vaznaugh et al., 2015

⁸⁵ Sanchez, 2021

⁸⁶ Springer et al., 2015

⁸⁷ Brunner & Vincent, 2018

⁸⁸ Richards, 2020

⁸⁹ Heath et al., 2022

⁹⁰ Meier et al., 2018

⁹¹ Weininger et al., 2015

⁹² Lance & Kachel, 2024

⁹³ Lapan et al., 2012

⁹⁴ Hemelt et al., 2021

Measure	Bottom-Third Country	Middle-Third Country	High-Income Country
Instructional resources: The availability, format, and quality of educational materials, including textbooks, digital tools, and Open Educational Resources (OER).	Students face restricted access to updated textbooks, digital tools, and library resources. ^{95 96}	Students benefit from basic instructional materials and occasional access to digital resources.	Students regularly use high-quality instructional materials, with strong access to books, technology, and digital tools. ^{97 98}

⁹⁵ Mumma, 2024

⁹⁶ Reardon, 2011

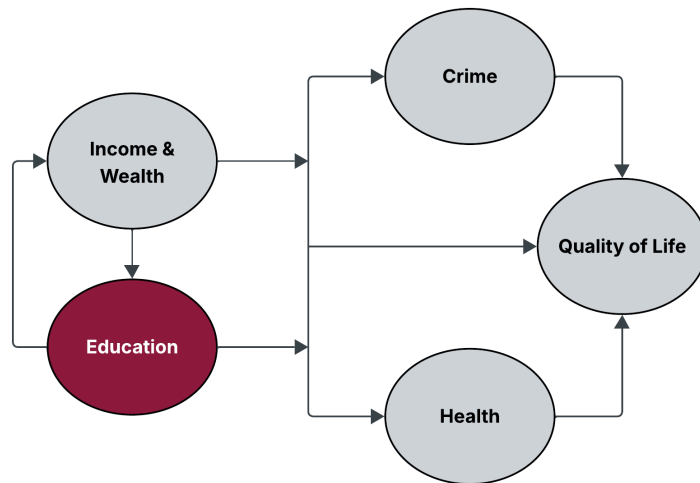
⁹⁷ Mumma, 2024

⁹⁸ Reardon, 2011

Appendix B: One-Third Countries Project Structure

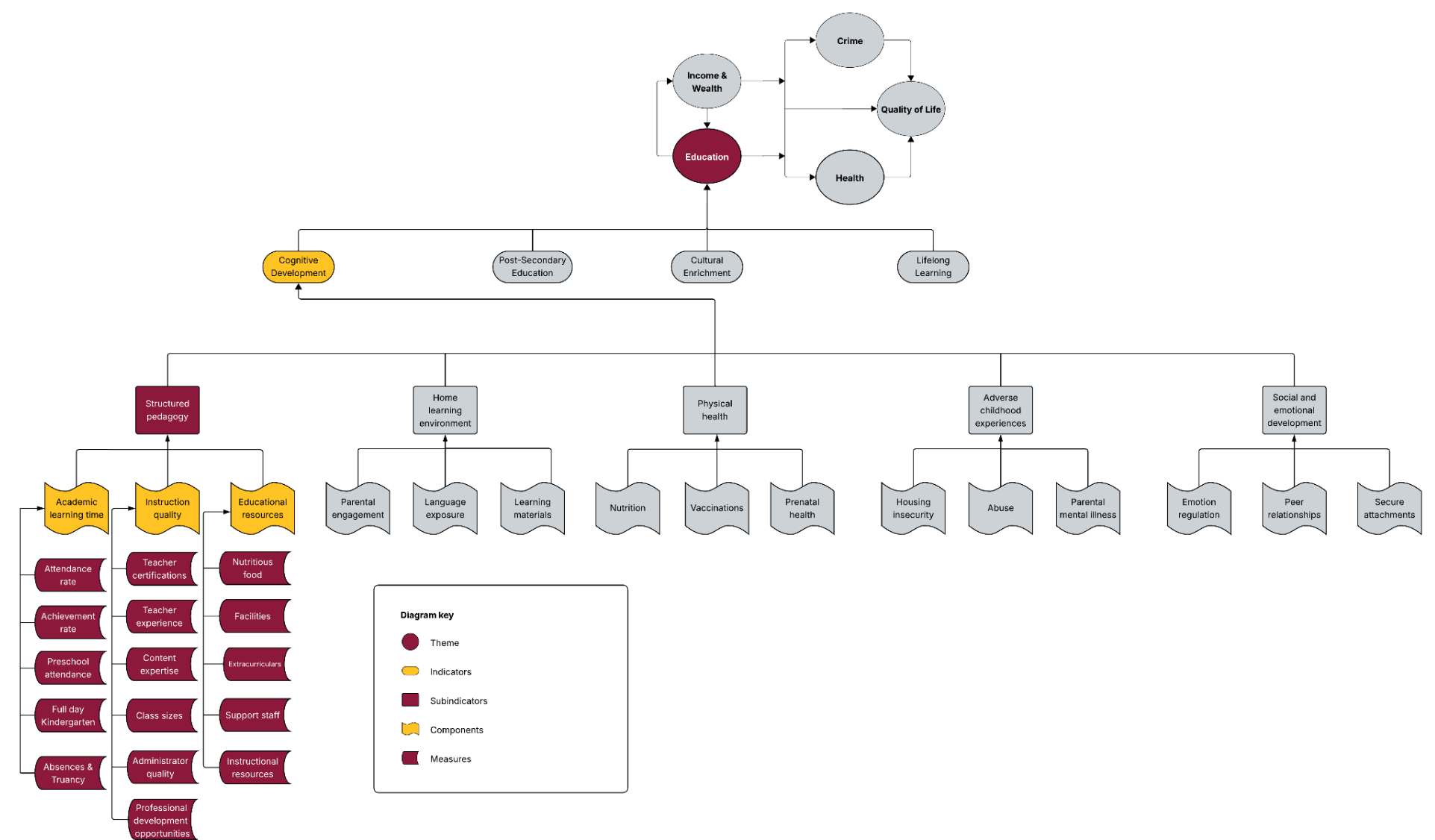
The overall narrative structure of the One-Third Countries Project is shown in Figure B1. The main focus of the narrative is to highlight income and wealth as a key predictor of outcomes in education, crime, and health, significantly impacting overall quality of life. At the same time, education has the potential to serve as the primary vehicle to “bulldoze” the borders between the three countries, unlocking the potential of each individual, despite their country of origin.

Figure B1: One-Third Countries Project Thematic Structure



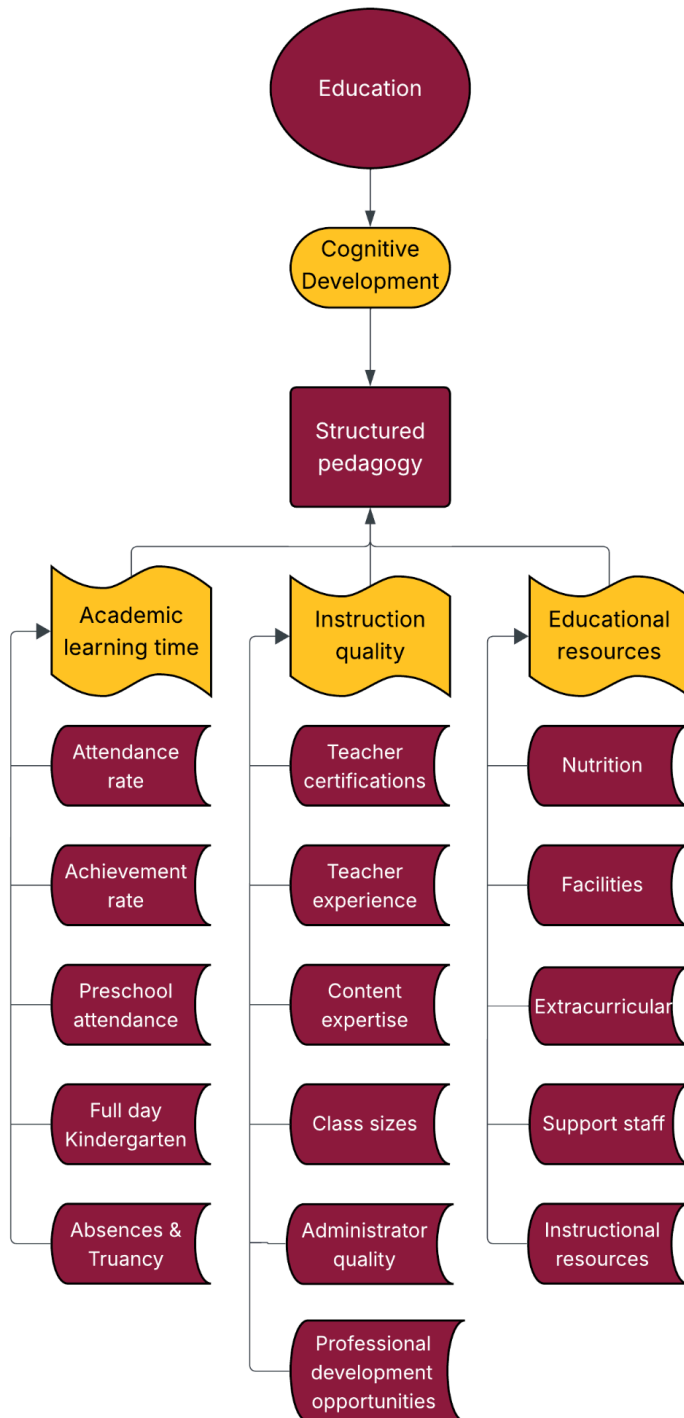
The narrative within each of the themes identified in Figure B1 will be structured using a hierarchical relational design of indicators, sub-indicators, components, and measures. Figure B2 displays a prototype for the education thematic hierarchy.

Figure B2: Education Hierarchical Narrative Design



Finally, Figure B3 displays the design that served as the structure for this narrative.

Figure B3: Hierarchical Narrative Structure for Structured Pedagogy



Appendix C: Detailed Narrative of Academic Learning Time

The initial narratives produced for this project were much longer, necessitating a substantial reduction of content. One of these longer narratives, covering academic learning time, is presented below.

Bottom-Third Country

For children in the Bottom-Third Country, the school day often begins with an invisible deficit. Truancy remains high among low-income students,⁹⁹ with empty chairs a result of external pressures,¹⁰⁰ degrading school facilities,¹⁰¹ and the quality of instruction itself.¹⁰² This loss of effective instructional time reduces the efficacy of policies aimed at closing achievement gaps,¹⁰³ while promulgating those same inequities in student outcomes.¹⁰⁴

Attendance rates are an issue in this tier. Chronic absenteeism, which is defined as a student missing at least ten percent of enrolled school days or more than ten days a month, is prevalent.¹⁰⁵ Students eligible for free lunches are twice as likely to be chronically absent,¹⁰⁶ demonstrating that students from lower-income brackets tend to have higher rates of chronic absenteeism across all ethnic groups.¹⁰⁷ This means low-income students are less able to benefit from sequential, cumulative learning experiences. The lack of exposure to consistent, structured learning experiences results in difficulty maintaining cognitive momentum for students.

Achievement rates and educational outcomes also vary distinctly by socioeconomic status. For example, only 7% of low-income students ascend to “high-achieving” status by 12th grade, and 44% of low-income students who start as high achievers in early grades fall out of the top performance quartile over time due to barriers such as limited access to rigorous coursework and support.¹⁰⁸ Additionally, low-SES students enter high school with literacy skills up to five years behind their high-income peers, face an 11.6% dropout rate between ages 16–24, and are significantly less likely to complete college.¹⁰⁹ These disadvantages for students in the bottom third have lifelong implications when it comes to future opportunities limiting their access to higher education, well-paying jobs, and social mobility, while also perpetuating cycles of inequality across generations.

⁹⁹ Black & Elgaddal, 2024

¹⁰⁰ Kelly, 2008

¹⁰¹ Kurtz, 2014

¹⁰² Fisher et al., 2015

¹⁰³ Wyner et al., 2007

¹⁰⁴ Kelly, 2008

¹⁰⁵ Garcia & Weiss, 2018

¹⁰⁶ Ibid, 2018

¹⁰⁷ Oregon Department of Education, 2015

¹⁰⁸ Wyner, et al., 2007

¹⁰⁹ American Psychological Association, 2017

Maximizing instructional time and ensuring an effective classroom environment are both critical for student learning and engagement. Students in the bottom income tier experience classrooms where teachers find themselves dedicating more class time to combat disruption- ultimately preventing seamless or well-utilized instructional time. The tempo of learning is slowed within low SES classrooms experiencing more time dealing with disciplinary actions and interruptions.¹¹⁰ In addition to lost time combating disruption, students in high-poverty high schools lose significant instructional time due to external community stressors and poor school conditions.¹¹¹ An estimated 18 minutes per day are lost due to delays, routines, or interruptions in high-poverty schools, and these schools also tend to lose 12.4% of their total instructional days throughout the school year due to teacher absences.¹¹² This inconsistent rhythm makes sustained student engagement cumbersome as student focus is often taxed by the surrounding environment- rather than by the academic material itself.¹¹³ Students show low to moderate engagement, with less consistent participation and reduced attentiveness during classroom activities.¹¹⁴

In this setting of fractured instructional time, the structure of effective teaching begins to break down. Student engagement plummets in the midst of a chaotic, overcrowded classroom environment, especially when students spend two-thirds or more of their time seated¹¹⁵ with minimal instructional interaction. In this vein, instructors struggle to complete lesson plans and address individual student shortfalls under the burden of consistent disruption.¹¹⁶ Furthermore, with students at widely different levels of understanding due to accumulated learning gaps, it is common for tasks to be either too easy or too difficult. A low academic achievement rate is correlated with a misdiagnosis of student skill levels,¹¹⁷ demonstrating how the lack of individualized instruction in the Bottom-Third reinforces these achievement gaps.¹¹⁸ In short, the very system designed to build knowledge is strained by the simple lack of consistent, protected time.

The long-term effect of lost instructional time is a significant impediment to cognitive development. Research indicates that differences in ALT caused by income begin before kindergarten and accumulate over time.¹¹⁹ The long-term effects include persistent disparities in reading, math, language, and executive function by third grade and beyond.¹²⁰ Ultimately, a child's mind requires consistent and successful engagement with new ideas to build strong cognitive frameworks. When that engagement is constantly interrupted and incomplete, development in foundational areas can be slowed. This systemic disadvantage impacts the development of higher-order skills like critical thinking and problem solving, creating barriers to future opportunities.

¹¹⁰ Rogers et al., 2014

¹¹¹ Ibid, 2014

¹¹² Ibid, 2014

¹¹³ Fisher et al., 2015

¹¹⁴ Kelly, 2008

¹¹⁵ Fisher et al., 2015

¹¹⁶ Kelly, 2008

¹¹⁷ Fisher et al., 2015

¹¹⁸ Kelly, 2008

¹¹⁹ Rakesh et al., 2025

¹²⁰ Ibid, 2025

Middle-Third Country

For students in the Middle-Third Country, the utilization and quality of academic learning time is generally consistent, to the substantial benefit of learning outcomes. Attendance rates are more steady in this tier than the one prior, with rates of truancy being low among middle-income students, and attending students' routines being consistent.¹²¹ Middle-income students have greater annual instructional hours, mostly unencumbered by classroom disruption or external pressures. The classroom environment that middle-income students inhabit is attentive to student weaknesses and responsive to requests for teacher intervention, in cases where further instruction is necessitated. Students in this income tier tend to engage by seeking additional help in class, which leads to increased support and teacher interaction, shaping a more responsive climate and increasing the quality of learning time.¹²² Though this learning paradigm strengthens teacher-student relationships and furthers academic achievement; it's a reactive model, which might diminish the frequency of opportunities for development beyond baseline competency, or fail to identify gaps unprompted.

When measuring academic achievement, middle-income students perform adequately as a result of strong safety nets promoting foundational skills and preventing egregious skill gaps. Middle-income student achievement has risen recently, yet continues to serve as a national midpoint for academic success. Middle-SES students' performance and dropout rates (3.6%) align closely with national averages, with academic achievement consistently ranking between low and high SES students. Access to resources, advanced coursework, and student support services is generally better than low-SES peers but lags high-SES schools.¹²³ This propensity towards adequacy improves graduation rates, yet diminishes the likelihood of students pursuing further education, especially at elite institutions; while simultaneously suggesting an overly rigid learning structure where the typical lesson-plan is not wholly reflective of presently measured or desired proficiencies. Students show moderate engagement, regularly participating and staying on task for most of the lesson.¹²⁴

Top-Third Country

The Top-Third Country tier is characterized by a highly resourced, high-income educational system. The educational model exhibited in the Top Third Tier is predicated on the idea that learning outcomes are directly proportional to academic learning time. Unlike in the Bottom-Middle-Third Countries, where socioeconomic factors lead to cumulative disadvantage and unequal access to structured learning, the Top-Third Country systematically eliminates barriers to academic learning time. Resources, culture, and policy are aligned to ensure time is not merely allocated, but effectively utilized.

¹²¹ Garcia & Weiss, 2018

¹²² Schweig et al., 2020

¹²³ Bradley, 2022

¹²⁴ Jensen, 2013

In the Top-Third, students experience steady attendance with fewer disruptions across grade levels.¹²⁵ Such a high rate is not enforced through punitive measures, but reflects the culture among Top-Third students where education is the primary vehicle for maintaining social and economic standing. This cultural orientation fosters a sense of responsibility, with families and communities of students in the top third reinforcing the expectation that consistent attendance is non-negotiable. As a result, students often approach school not merely as a requirement but as a guaranteed pathway to long-term success, which enhances their intrinsic motivation, engagement in learning, and persistence through challenges. In comparison, nationally representative data shows that students eligible for free and reduced lunch are twice as likely to miss 10 or more days a year when compared to higher-income peers.¹²⁶ Additionally, chronic absenteeism is 16% as prevalent in high-poverty U.S. schools, compared to a mere 8% in low-poverty schools.¹²⁷ For students, consistent attendance and the absence of chronic absenteeism mean that instruction rarely needs to be reviewed for missing students, maximizing allocated time for actual teaching. This continuity in attendance is critical for cognitive development, as repeated exposure to cumulative, sequenced content allows students to build cumulative skills in literacy, numeracy, and executive function. Lessened rates of chronic absenteeism is another benefit to the top third income level as chronic absenteeism leads to increased grade retention, reduced graduation rates, and lower test scores/ chances of meeting state standards—lagging about 16 percentage points behind.¹²⁸

Instructional time in the Top-Third is focused, going beyond the standard school day with enrichment programs to maximize total learning hours, a strategy strongly associated with improved achievement. The system embraces expanded learning time by design, led by certified, field-leading instructors. While U.S. public schools offer similar scheduled instructional time across income levels, a 200-hour gap exists between schools at the top and bottom percentiles.¹²⁹ Although U.S. public schools schedule similar amounts of instructional time, this 200-hour gap reflects how students in the top third actually gain more usable learning time because their schools face fewer interruptions from factors like absenteeism, discipline issues, staffing shortages, or limited resources. For high-income students, this additional time compounds over years of schooling, giving them the advantage of extra learning that allows for mastery of core subjects and additional space for enrichment activities that build college readiness and expand on critical skills. Meanwhile, though schools in the Bottom-Third Country often schedule longer days, they lose 16–25 percent of that time due to absences, suspensions, and disruptions.¹³⁰ Students in high-poverty high schools, in comparison, receive an average of half an hour less instruction per day when compared to higher-income peers.¹³¹ Cognitively, more allocated, uninterrupted time towards instruction means students in this tier spend longer periods engaging in inquiry, mastering skills, and practicing critical thinking, which enhances memory consolidation and higher-order reasoning.

¹²⁵ Gennetian et al., 2018

¹²⁶ Garcia & Weiss, 2018

¹²⁷ The Commonwealth Institute, 2017

¹²⁸ Oregon Department of Education, 2015

¹²⁹ Kraft & Novicoff, 2022

¹³⁰ Kraft & Novicoff, 2022

¹³¹ Kurtz, 2014

The achievement rate, as a result, is uniformly high, with the system operating well beyond basic competency. Students tend to achieve more, supported by early advantages and well-resourced instruction.¹³² Academic outcomes vary significantly by socioeconomic status, with low-SES students entering highschools with literacy skills up to five years behind their high-income peers and facing a dropout rate of 11.6% compared to a high-SES student dropout rate of 2.8%—making high income students eight times more likely to earn a bachelor's degree by age 24.¹³³ Achievement is strongly correlated with the maximized ALT, as students consistently work on appropriately prescribed learning tasks that ensure a high success rate. Teachers are adept at diagnosing skill levels and ensuring content coverage aligns precisely with assessment goals, tailoring material to a student's current skill level. As a result, students master core subjects earlier, which frees cognitive resources for more advanced learning tasks and enrichment opportunities. In these environments, differentiated instruction is embraced to emphasize critical thinking, project-based learning, and student-driven inquiry, giving students opportunities to develop deeper conceptual understandings of the material and resilience in tackling complex material. There is also some correlation between household income and student test scores, with one study finding that a \$10,000 increase in income is correlated with a 3.7 point score increase on standardized testing,¹³⁴ meaning students in this income tier are more likely to meet or excel state standards.

Classroom environments in the Top-Third are characterized by low student-to-teacher ratios, strong routines, and effective management. These environments are dedicated to substantive interaction between the students and their instructors, particularly in the monitoring and provision of feedback on academic material. Students receive frequent individualized attention and benefit from proactive teacher interaction, fostering a highly responsive and structured classroom climate.¹³⁵ The highly responsive and structured classroom climate with proactive teacher intervention fosters a favorable disciplinary climate, supporting students' ability to focus and work well. Students' ability to focus is crucial for maximizing working memory during complex tasks, directly enhancing cognitive development.

Additionally, students are buoyed by their successes and are motivated to increase their engagement and effort, which creates a positive feedback loop for cognitive growth. Most students in this tier remain actively on task, with sustained attention and consistent participation.¹³⁶ Structured pedagogy in this tier is focused on fostering constant dialogue and interaction. The clarity of expectations and breaking down of complex tasks help reduce cognitive overload, making participation less intimidating and more consistent, leading to better student outcomes.

¹³² American Psychological Association, 2017

¹³³ American Psychological Association, 2017

¹³⁴ Sheu & Dada, n.d.

¹³⁵ Boyce et al., 2012

¹³⁶ Bae & Lai, 2020

Appendix D: Technical Methodology for Visualizations

The visualizations presented in this report were produced using a modular analytical framework developed for the 1/3 Country Project. The methodology ensures that diverse datasets can be analyzed and visualized consistently against a standardized income distribution. The process involves four main stages:

1. Estimating household income for geographic areas,
2. Applying the 1/3 Country technical framework,
3. Using standardized templates for data preparation, and
4. Generating the final visualizations.

1. Geographic Area Household Income Estimator Tool

A key component of this project is a comprehensive database that provides reliable median household income estimates for a wide range of U.S. geographic areas, such as school districts, census tracts, and zip codes. This resource, referred to as the *Geographic Area Household Income Estimator Tool*, was constructed to assign an income level to administrative datasets that lack their own income information.

The tool was built using a multi-step process:

- **Data aggregation:** Publicly available summary tables from the IPUMS National Historical Geographic Information System (NHGIS), containing published median incomes and income distributions for various geographic levels, were collected. This was supplemented with detailed, person-level microdata from the IPUMS American Community Survey (ACS), which includes continuous household income and statistical weights.
- **Advanced income estimation:** For geographic areas where the published median income was missing, suppressed, or otherwise unreliable, an advanced estimation technique was employed.
 - a. First, the income distribution data for the specific area (e.g., a school district) was used to identify the income bracket containing the 50th percentile.
 - b. Next, the more detailed ACS microdata was used to calculate the precise weighted median income of all households falling within that specific income bracket and larger parent geography (the Public Use Microdata Area).
 - c. This calculated value was then used as the estimated median income for the smaller geographic area.
- **Geographic harmonization:** To ensure accuracy across different time periods and boundary changes, geographic boundary files (shapefiles) from before and after the 2020 Census were downloaded from the U.S. Census Bureau. Using official relationship files, a crosswalk was created to link older and newer geographies. This allowed for the precise spatial assignment of smaller areas to their correct parent geographies, which is critical for accurate income estimation.

The final output is a master database that serves as a single source of truth, allowing databases utilizing census tracts, public use microdata areas (PUMAs), zip codes, school districts, and urban areas to be joined with a high-quality income estimate.

2. The 1/3 Country Technical Framework

The core of this project is a technical framework designed to analyze socioeconomic indicators by situating them within the income distribution of the United States. This technical framework was developed during Phase 1 of the 1/3 Countries Project.

- **Conceptual foundation:** The framework reimagines the U.S. population as three distinct "countries" based on household income: the bottom third, middle third, and top third. For more granular analysis, each of these terciles is further subdivided into 20 ventiles, creating 60 consecutive, fine-grained income groups.
- **Standardized income borders:** A foundational step was the one-time calculation of definitive income boundaries for each of the 60 groups. Using a large, nationally representative, and weighted sample from the American Community Survey (ACS), a master set of income thresholds was generated. This ensures that a specific income level is classified identically across all analyses, regardless of the original data source, thereby ensuring consistency and comparability between different indicators.
- **Modular functions:** The framework is built on a library of shared, reusable code. This includes utility functions that can take any dataset and consistently assign each observation (e.g., a person or a school district) to its correct income group based on the master thresholds. It also includes a suite of standardized plotting functions that generate all visualizations with a uniform aesthetic, including the three colored backgrounds that represent the income "countries."

3. Creation of Templates for Standardized Analysis

To enable the rapid and consistent analysis of new indicators, a series of "plug-and-play" script templates were developed.

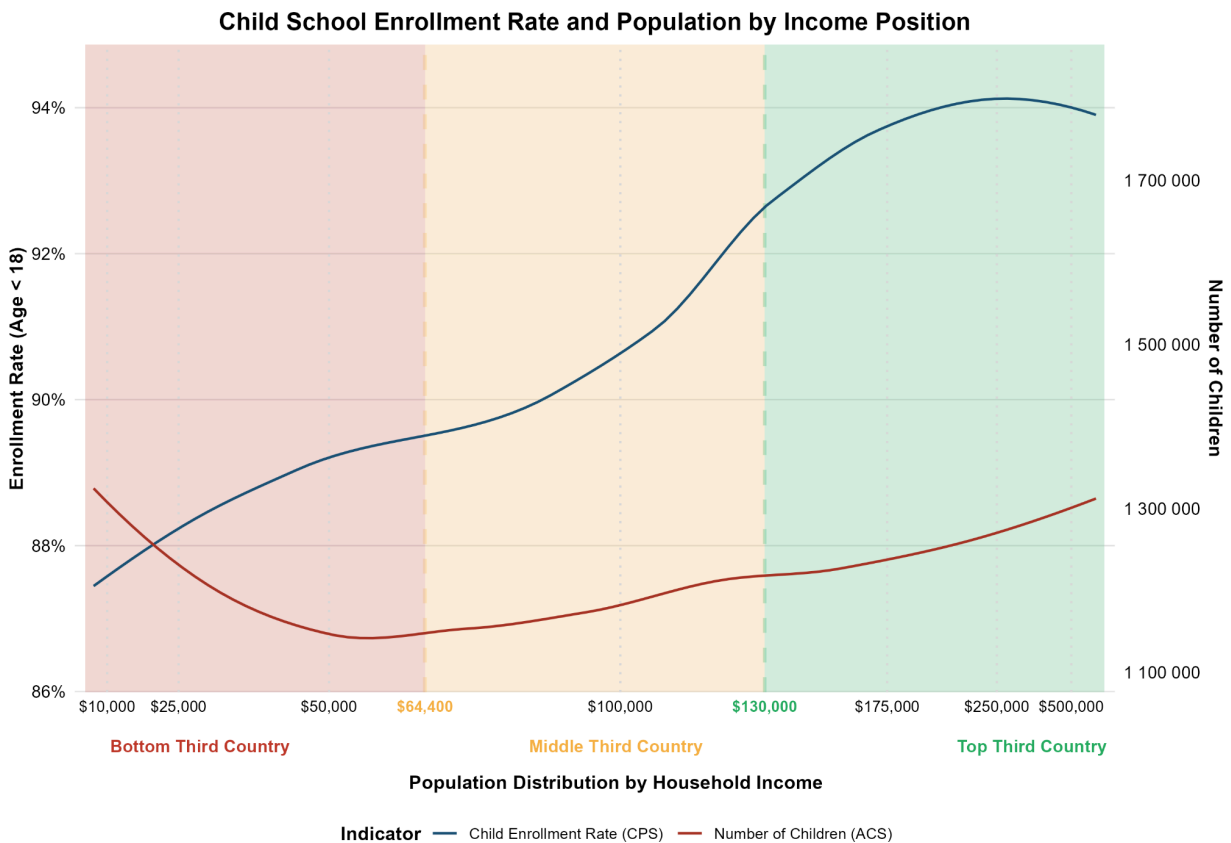
- **Purpose and design:** The goal of the templates is to streamline the process of acquiring, cleaning, and preparing new data for visualization while ensuring adherence to the project's standards. Each template is a pre-structured script with clearly delineated sections for user inputs (e.g., data file paths), data cleaning logic, the calculation of the specific indicator, and a final call to the shared visualization functions.
- **Workflow:** This modular design separates data-specific logic from the core framework. A researcher using the system simply copies a template, provides the necessary inputs in the designated section, adds a few lines of code to clean their data and calculate their specific variable of interest, and executes the script. This process produces a finalized analysis and a standardized visualization that is immediately comparable to all others in the project.

Templates were then created for the Integrated Public Use Microdata Sample (IPUMS) ACS, IPUMS Current Population Survey (CPS) (including the Annual Social and Economic Supplement (ASEC), Voting Supplement, and Education Supplement), and Urban Institute Education Data Portal (UIEDP).

4. Creation of the Report Visualizations

Each visualization in this report was generated by applying the framework and templates described above.

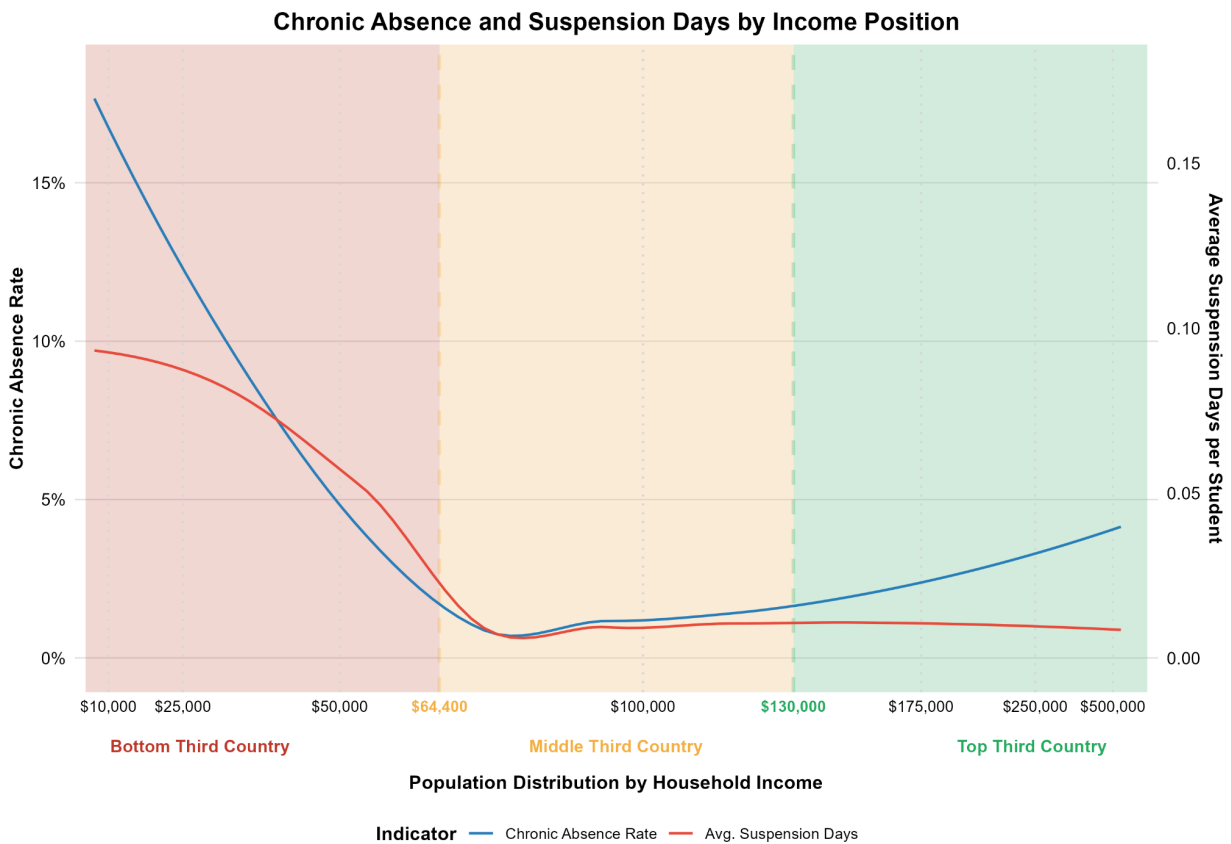
Figure D1: Child School Enrollment Rate and Population by Income Position



This dual-axis plot combines data from two different sources. The child enrollment rate was calculated from the IPUMS CPS October Education Supplement, while the child population count was calculated from the IPUMS ACS. Because the CPS dataset contained income only in broad categories, a continuous income value was statistically imputed for each household by performing a weighted random draw from the corresponding income bracket in a donor dataset (the CPS Annual Social and Economic Supplement). After assigning individuals from both

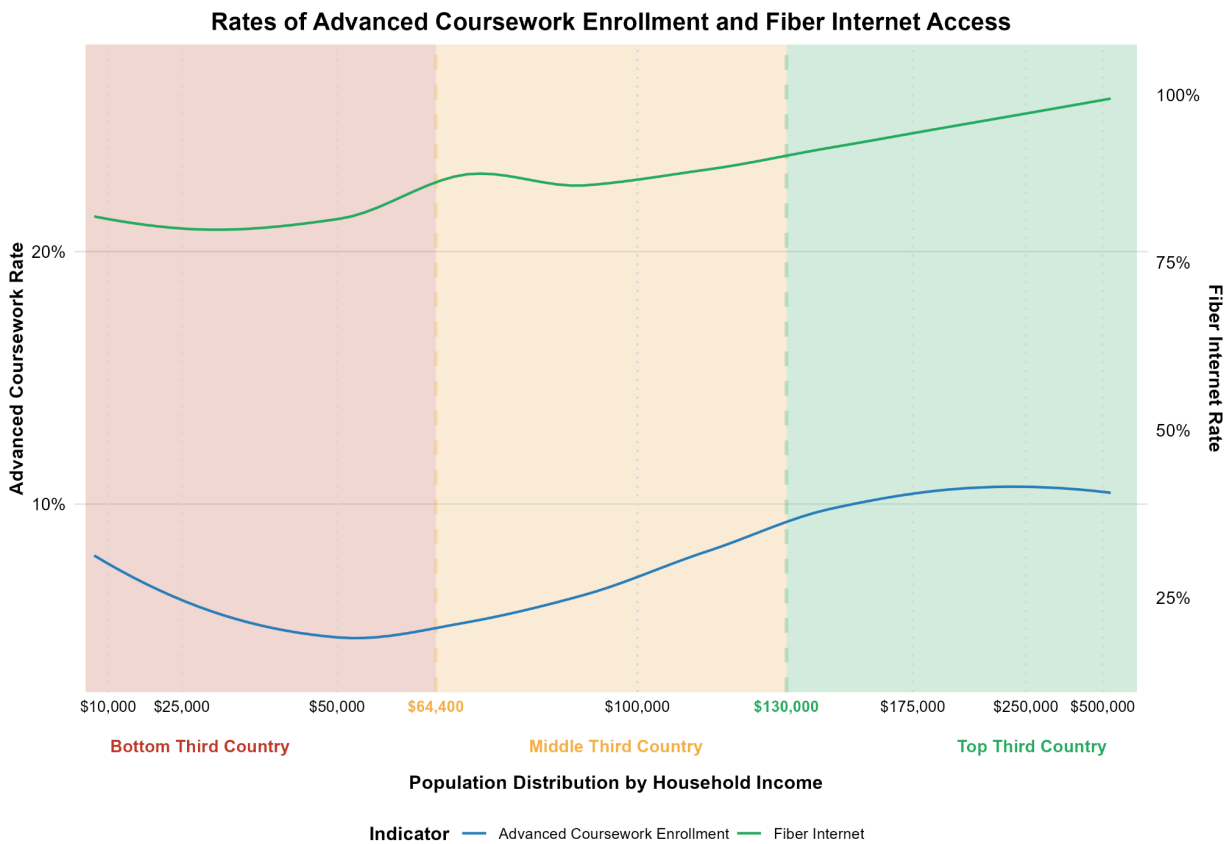
datasets to their respective income groups, summary statistics were calculated for each of the 60 ventiles. Finally, the two summary tables were merged and visualized.

Figure D2: Chronic Absenteeism and Suspension Days by Income Position



This visualization used data from UIEDP for the 2017 school year. School-level data on enrollment, suspensions, and absenteeism were first aggregated to the administrative school district level. Each district was then assigned an estimated median household income using the Geographic Area Household Income Estimator Tool. The final indicators—chronic absenteeism rate and average suspension days per student—were calculated for each district. For the visualization, districts were sorted into the 60 income ventiles, and a weighted average of each indicator (using district enrollment as the weight) was calculated for each group and plotted.

Figure D3: Rates of Advanced Coursework Enrollment and Fiber Internet Access



This visualization also used data from UIEDP, combining data across different years to use the most recent information available for each metric. Data for advanced coursework and enrollment were from the 2017 school year, while data for fiber internet access was from the 2020 school year. The preparation process mirrored that of Figure C2 was merged and aggregated to the district level, each district was assigned an estimated median income, and the final indicators were calculated. The fiber internet rate represents the proportion of schools in a district with fiber access, while the advanced coursework rate is an average of the district's enrollment rates in AP, IB, and Gifted & Talented programs. The final plot shows the enrollment-weighted average of these rates for each of the 60 income ventiles.

A list of software tools utilized in this project is available at the end of the References section.

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